

Using AI for Study and Data Analysis

Quantitative Reasoning II (INF 1220)

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1. Introduction

In this course, we use LLMs and coding agents. LLMs are powerful tools for studying, coding, and other tasks. Understanding how to direct an LLM is a skill that takes practice. This technology is likely here to stay, and no matter your future plans, understanding its capabilities is important.

This document is my first attempt at distilling some of the best practices I have learned over the years that are relevant to this course. Consider it a starting point. My practices (and yours) will necessarily evolve along with the technology. This document applies to two major types of AI use: study and agentic coding for data analysis. The final section applies specifically to how you will use LLMs to evaluate your own work in this course.

I use “LLM study” to refer to engagement of an LLM to learn a new topic. Effective LLM-assisted study and research requires *proactive engagement*. This is true for all scholarly efforts, but especially so for LLM-assisted study. A standard textbook provides a structured presentation of material and does not hallucinate. An LLM makes errors with some frequency (and does not change its tone when it does), assume background knowledge you do not have, or present information that is not relevant to the topic at hand. You must learn to apply critical thinking, creativity, and curiosity to your interactions with LLMs.

Agentic coding refers to the use of an LLM to generate code for a specific task. In this class, you are not required to learn how to write code (though you are welcome to use these assignments as an opportunity to do so). Agentic coding allows you to focus your time and attention on the *design* of your analysis, rather than the implementation. Producing a data analysis requires a number of careful judgments that you must make (and be able to explain) yourself.

A couple of analogies may be helpful: when studying, think of the model as a more experienced classmate (knowledgeable, but not infallible). When coding, think of the model as an assistant or intern: you must communicate carefully what you want your assistant to create. If instructions are not clear, the model will make decisions for you that you may not agree with.

2. Engage proactively

Asking a question and getting an answer is the first step. An effective conversation includes follow-up questions, or, if data analysis code is involved, extensions or improvements to the code in question. In LLM study, each topic conversation must end with a succinct rephrasing of what you learned in your own terms.

Note: LLMs often offer suggested follow-up questions or lead you in a particular direction. I prefer to not only ignore these suggestions, but ask the model not to offer such questions (this can be done in your initial prompts or in your account settings) because they encourage passive engagement.

3. Bring something to the table

A reasonable rule of thumb for these interactions: A sophisticated conversation with an LLM will include a point at which you identify an error, clarify a point, or uncover an omission. If your conversation contains no such point, one of two cases applies. Either you worked so shallowly that the model could have handled your half of the exchange (see my note about LLM-offered follow-up questions), or it made an error, omission, or over-simplification and you accepted the inadequacy without questioning it. Importantly, models (like humans) can behave sycophantically: you can get the model to agree with you even when you are wrong. One way to combat this is to explicitly ask the model to disagree with you when you are wrong. Another is to leverage fresh chat sessions to confirm your own understanding. Bottom line: Maintain professional skepticism even when the model sounds helpful and confident.

4. Have it test you

A good text offers review questions and exercises. An LLM can do the same. For each topic you study, ask the LLM to provide a test of 3-5 questions for you to answer, and to evaluate your answers. Ask for questions more than just multiple-choice. Here is a sample prompt, which you may adjust per topic:

"Please give me a quiz of 4 questions on the *<fill in>* subject we just discussed and which I just summarized. Prefer questions that ask me to generalize on or extend the ideas, not just recite facts or vocabulary. Include at least one question that asks me to extend on or creatively modify the code we discussed."

5. Evaluate the conversation

Finish each LLM assignment with the prompt below, pasted as written, with the assignment's list of topics or tasks filled in at the end. Turn in the conversation.

I will read your conversation myself and assign the final grade. The model's evaluation is the baseline for that grade, not the grade itself.

From the perspective of the instructor of an undergraduate statistics and data science course, review this entire conversation and evaluate it. Be frank, terse, and constructively critical. Identify specific missed opportunities. Do not give high scores for merely adequate work.

Specifically, for each of the topics listed below, *topic by topic*, evaluate these points:

- A. Did we engage in a deep conversation with thoughtful follow-up questions from me? Were there significant follow-up questions I might have asked but missed? Cite examples where relevant.
- B. When there was relevant code, did we work together to improve or add to that code?
- C. Did I summarize the topic accurately?
- D. Did I get a quiz from you on the topic, and how did I do on that quiz?
- E. Did I at any point catch you in a significant error which you had to correct, or find a way to notably improve or refine your responses?
- F. Give our conversation a 5-point score using this rubric:
 - 1. **(Lowest)** The conversation was largely a glorified websearch. There were few if any follow-up questions, the quiz was nonexistent or I got most answers wrong, and my summaries were inaccurate or missing.
 - 2. I at least made some effort to follow up, summarize, and answered some quiz questions correctly, but it was still below average.
 - 3. I followed up, summarized and got most quiz questions correct, but missed interesting follow-up possibilities and some of the quiz questions.
 - 4. I did all that would reasonably be expected in following up on questions, summarized well, and got almost all quiz questions right.
 - 5. **(Highest)** I did all that is expected under 4, and I also either caught and corrected an error on your part, or offered a significant improvement or refinement to a point you made. If we had been two people talking, you would say that you also learned something from me.

<List of Topics, varying per assignment>

6. Turning in a conversation

Major LLMs have a "share conversation" feature. When applicable in the homework assignment, paste the link to the conversation into the Populi assignment submission. Before submitting, open your own link in a private browsing window to check that the sharing settings are correct (anyone with the link can view).

Important: Sharing a chat shares the whole conversation, so do not put anything private into a chat you intend to turn in.